



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Agile Project Management for Business

TGS Ref No: TGS-2023018967

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version v9.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
v9.0	17 August 2026	Full content revamp. Rebuilt on a single-source content module. Expanded Agile theory (Manifesto values/principles, Scrum accountabilities/artefacts/events, Lean, Kanban + Little's Law, XP, DSDM, scaling). Added the five Agile metrics with worked interpretation. Replaced generic exercises with 8 tool-based activities on one running case study (HarbourFront Logistics). Added activity debriefs and full TSC mapping.	Dr Alfred Ang
v8.0	28 November 2022	Previous released version.	Tertiary Infotech Academy

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1. Course Information

Field	Detail
Course Title	Agile Project Management for Business
WSQ Course Code	TGS-2023018967
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN: 201200696W)
Course Duration	2 days · 16 instructional hours (8 hours per day) · plus 2 hours assessment
Delivery Mode	Physical classroom, synchronous Zoom, or corporate on-site delivery
Class Size	Maximum 20 learners per class
TSC Reference	Business Agility · ICT-BIN-4038-1.1
Assessment	Written Assessment (WA — SAQ, 1 hour) and Case Study (CS, 1 hour). Both open book.
Trainer	Dr Alfred Ang
Version	v9.0 · 17 August 2026
Course Page	https://www.tertiarycourses.com.sg/wsq-agile-project-management-for-business.html

1.1 Target Audience

This course is designed for professionals who lead, coordinate or contribute to projects and who need to deliver in conditions where requirements change. Suitable job roles include:

- Project Manager
- Scrum Master
- Product Owner
- Business Analyst
- Team Leader
- IT Project Manager
- Change Manager
- Operations Manager
- Programme Manager

1.2 Entry Requirements

- Basic computer operation proficiency
- Minimum 3 GCE 'O' Level passes including English, or Workplace Literacy (WPL) Level 5
- Minimum 1 year of working experience
- A positive learning attitude and willingness to participate in team activities

1.3 Learning Outcomes

By the end of this course, learners will be able to:

- LO1: Adopt a new Agile mindset for project management
- LO2: Share and implement Agile practices within the teams
- LO3: Build an Agile team to execute and track the project performance

1.4 TSC Mapping — Abilities and Knowledge Assessed

All content maps to the Skills Framework TSC “Business Agility” (ICT-BIN-4038-1.1). Nothing is assessed that is not taught and practised.

Code	TSC Ability	Taught in
A1	Share information actively within and across teams to bridge operational barriers	Topic 2 · Activity 4
A2	Organise work in alignment with operational priorities	Topics 1–2 · Activities 2, 4
A3	Implement Agile or lean practices to reduce waste and defects in operating procedures and practices	Topics 2–3 · Activities 3, 7
A4	Measure progress against targets for defined business outcomes on a regular basis	Topic 3 · Activities 5, 7, 8
A5	Experiment with new ideas, products or services	Topic 1 · Activity 1
A6	Assess work performance and quality to ensure continuous improvement	Topic 3 · Activities 6, 7, 8
A7	Manage individual work responsibilities and take ownership of individual and team outcomes	Topic 3 · Activities 5, 6, 8

Code	TSC Knowledge	Taught in
K1	Methods to analyse current and future business operating landscapes	Topic 1 · Activity 2
K2	Methods to analyse current and future customer needs and preferences	Topic 1 · Activity 1
K3	Organisational policies, processes and standards	Topic 1 · Activity 2
K4	Types of change management methodologies, tools and practices	Topic 2
K5	Types of team composition and formation models	Topic 3 · Activities 4, 5
K6	Values and principles of Agile methodologies	Topic 2 · Activity 3
K7	Types of Agile methodologies and practices	Topic 2 · Activity 3

2. Training Resources

2.1 Trainer Resources

- Master Trainer Slides (PPTX and PDF), 120 slides, version v9.0
- This Lesson Plan
- Learner Guide with full step-by-step activity instructions
- Assessment instruments: WA (SAQ) question paper and answer key; Case Study question paper and marking guide (trainer-only — never distributed to learners)
- Laptop, projector and internet access
- Whiteboard and markers
- Sticky notes and A3 flip-chart paper for the team activities
- Planning-poker cards (Fibonacci: 1, 2, 3, 5, 8, 13, 21) — one deck per team

2.2 Learner Resources

- Learner Slides (PDF)
- Learner Guide (PDF) — contains every activity step
- A laptop or tablet with a modern web browser and internet access
- The HarbourFront Logistics case brief (in the Learner Guide)

2.3 Online Tools Used in the Activities

All tools are browser-based. Nothing needs to be installed and no login is required.

Tool	URL	Used in
5 Whys	https://alfredang.github.io/5whys/	Root-cause analysis in the sprint retrospective (Activity 6)
Fishbone	https://alfredang.github.io/fishbone/	Cause-and-effect analysis of delivery problems (Activity 2)
Pareto Chart	https://alfredang.github.io/paretochart/	Prioritising the 20% of causes driving 80% of defects (Activity 7)
RACI Matrix	https://alfredang.github.io/raci/	Clarifying Agile role accountability (Activity 4)
Scrum Board	https://alfredang.github.io/scrum/	Running the sprint board, backlog and burndown (Activities 3, 5, 8)
Design Thinking	https://alfredang.github.io/designthinking/	Empathising with customers to shape the product vision (Activity 1)

3. Daily Training Schedule

Each training day runs 09:30–18:30 and delivers exactly 8 instructional hours. The 1-hour lunch break is excluded from instructional time; short tea breaks are counted within it. Slide references are to the Master Trainer Slides version v9.0.

3.1 Day 1 — The Agile mindset and the essentials

Instructional hours: 8.00 (excludes the 1-hour lunch break).

Time	Hrs	Topic / Activity	Content and Delivery Notes	Method	Slides
09:30 – 09:45	0.25	Digital Attendance (AM) & Welcome	Trainer displays the SSG digital attendance QR code; learners scan and submit. Housekeeping and safety brief.	Trainer-led Admin	2–7
09:45 – 10:15	0.50	Trainer & Learner Introductions · Ground Rules	Trainer introduction. Learners share role, industry and their biggest project frustration — captured on the whiteboard and referred back to all day.	Discussion Ice-breaker	4–6
10:15 – 10:45	0.50	Learning Outcomes · TSC Mapping · Course Outline	The three learning outcomes, how the TSC abilities and knowledge map to each topic, the six tools, and the running	Trainer-led Q&A	8–16

			HarbourFront case study.		
10:45 – 11:00	0.25	Assessment Briefing & Funding Criteria	Assessment format (WA + Case Study), open-book scope, the assessment flow, 75% attendance and TRAQOM requirements.	Trainer-led	17–19
11:00 – 12:15	1.25	Topic 1: Introduction to Agile Project Management (Part 1)	Are we building the right thing? The traditional project lifecycle and the challenges every PM recognises. The waterfall model, its assumptions, and why it struggles. The cost-of-change curve. (K1, K3)	Trainer-led Diagrams Discussion	20–26
12:15 – 13:00	0.75	Activity 1: Empathise with the Customer (start)	Teams of 3–4 use the Design Thinking tool to build an empathy map and persona for Priya Menon, then reframe the requirement as a customer-value problem statement. (K2, A5)	Hands-on Team activity Design Thinking tool	37–39
13:00 – 14:00	—	LUNCH BREAK (1 hour — not counted as instructional time)	Digital Attendance (PM) taken on return.	Break	44
14:00 – 14:30	0.50	Activity 1 debrief · Digital Attendance (PM)	Each team presents its reframed problem statement in 90 seconds. Trainer draws out the difference between signed-off scope and delivered value.	Presentation Debrief	39
14:30 – 15:45	1.25	Topic 1: Introduction to Agile Project Management (Part 2)	What Agile actually is. Agile vs traditional. Inverting the iron triangle. Methods to analyse the operating landscape (PESTLE, SWOT, Five Forces, scenario planning) and customer needs (personas, journey maps, JTBD, Kano, A/B). Agile beyond software. When	Trainer-led Comparison tables Case examples	27–36

			Agile is NOT the right answer. (K1, K2, A2, A5)		
15:45 – 16:30	0.75	Activity 2: Diagnose the Waterfall Failure (Fishbone)	Teams use the Fishbone tool to analyse why CustomerConnect failed, tag each cause structural or behavioural, and pair each structural cause with an Agile practice. (K1, K3, A2)	Hands-on Team activity Fishbone tool	40–42
16:30 – 16:45	0.25	Activity 2 debrief · Topic 1 recap	Why most causes tag as structural, and the two causes Agile does not fix on its own.	Debrief	42–43
16:45 – 18:15	1.50	Topic 2: Agile Essentials (Part 1)	The Agile Manifesto: 4 values read correctly and the 12 principles. The Scrum framework end to end — 3 accountabilities, 3 artefacts and their commitments, 5 events. The Definition of Done. (K6, K7)	Trainer-led Framework diagram Discussion	45–56
18:15 – 18:30	0.25	Day 1 recap and Q&A	Consolidate the mindset shift and preview Day 2.	Q&A	73–74

3.2 Day 2 — Building and running an Agile team

Instructional hours: 8.00 (excludes the 1-hour lunch break).

Time	Hrs	Topic / Activity	Content and Delivery Notes	Method	Slides
09:30 – 09:45	0.25	Digital Attendance (AM) & Day 1 recap	Learners scan the SSG QR code. Quick recall of the four values and the three Scrum accountabilities.	Admin Q&A	74
09:45 – 11:00	1.25	Topic 2: Agile Essentials (Part 2)	Lean and the seven principles, the eight wastes. Kanban's five practices and Little's Law. Extreme Programming. Choosing between Scrum, Kanban and Scrumban. DSDM/AgilePM and the scaling frameworks. (K7, A3)	Trainer-led Comparison table Board walkthrough	57–63
11:00 – 11:45	0.75	Topic 2: Change Management and	Finding Agile support: naming	Trainer-led Role discussion	64–65

		Resistance (K4)	and answering each stakeholder's actual fear. Making the adoption stick — willing pilot, visible wins, evolutionary change, coach then withdraw. The honest warning about 'more for less'. (K4, A1)		
11:45 – 12:45	1.00	Activity 3: Build the Product Backlog & Run Sprint 1 Planning	Teams write user stories with Given/When/Then criteria, test them against INVEST, estimate with planning poker, order with MoSCoW, write one sprint goal and load Sprint 1 to 20 points. Trainer acts as Product Owner. (K6, K7, A3)	Hands-on Team activity Scrum Board tool Planning poker	67–69
12:45 – 13:00	0.25	Activity 3 debrief	Why the sprint goal is the artefact that makes a sprint reviewable, and why splitting by workflow step beats splitting by technical layer.	Debrief	69
13:00 – 14:00	—	LUNCH BREAK (1 hour — not counted as instructional time)	Digital Attendance (PM) taken on return.	Break	—
14:00 – 14:45	0.75	Activity 4: Clarify Role Accountability (RACI) · Attendance (PM)	Teams map 12 project decisions against 5 roles, audit for duplicate/missing Accountables, and resolve the conflicting-priorities escalation. (K5, A1, A2)	Hands-on Team activity RACI tool	70–72
14:45 – 15:30	0.75	Topic 3: Building the Agile Team and Setting the Vision	Team composition and formation models, generalising specialists, Tuckman's stages and adaptive leadership, servant leadership, and setting a shared vision. (K5)	Trainer-led Diagrams	75–80
15:30 – 16:15	0.75	Topic 3: User Stories, Prioritisation and Estimation	Story anatomy, the three Cs, INVEST and the split rule. Prioritisation techniques. Relative	Trainer-led Worked examples	81–85

			estimation, story points, planning poker and the 13-point rule. Timeboxing and Parkinson's Law. (A2, A7)		
16:15 – 17:15	1.00	Activity 5: Execute Sprint 1 and Track It on the Board	Teams simulate all 10 sprint days with daily stand-ups, a WIP limit of 3, and three trainer-injected impediments. They plot the burndown and compute actual velocity. (K5, A4, A7)	Hands-on Simulation Scrum Board tool	99–101
17:15 – 17:45	0.50	Topic 3: The Five Agile Metrics, Waste and Risk	Sprint burndown, release burndown, velocity, control chart and cumulative flow diagram — what each shows, how to read it, and the anti-patterns. Little's Law worked numerically. Root-cause tools, the Pareto principle, value stream mapping, risk as anti-value, and technical debt and ownership. (A3, A4, A6, A7)	Trainer-led Chart interpretation	86–98
17:45 – 18:15	0.50	Activities 6, 7 & 8: Retrospective, Pareto and Forecast	Activity 6: full retrospective with 5 Whys to a changeable root cause and one SMART action. Activity 7: Pareto of 120 defects to find the vital few. Activity 8: velocity forecast as a range, and reading the CFD/control chart to name the bottleneck. (A3, A4, A6, A7)	Hands-on Team activities 5 Whys · Pareto · Scrum Board	102–110
18:15 – 18:30	0.25	Topic 3 recap · Course summary and next steps	Consolidate the three learning outcomes, the eight key takeaways, and the recommended follow-on courses. Final Q&A before the feedback survey.	Recap Q&A	111–115

3.3 Assessment Session (Day 2, after instructional hours)

The 2-hour assessment sits outside the 16 instructional hours.

Time	Hrs	Session	Notes	Method	Slides
18:30 – 18:45	—	Course Feedback and TRAQOM Survey (mandatory)	Learners complete the TRAQOM survey on lms-tms.tertiaryinfotech.com .	Admin	118
18:45 – 19:00	—	Digital Attendance (Assessment) and Assessment Briefing	Assessment attendance submitted. Rules restated: open book (slides, Learner Guide, approved materials only), no discussion, no photography.	Admin	116–117
19:00 – 20:00	—	Written Assessment (WA) — Short-Answer Questions	Individual, open book, 1 hour.	Assessment	116
20:00 – 21:00	—	Case Study (CS)	Individual, open book, 1 hour. An applied business scenario with structured tasks.	Assessment	116
21:00 – 21:15	—	Results, feedback and appeal process	Competent / Not Yet Competent communicated, with the appeal route explained.	Admin	117

4. Activity Summary

Every activity advances the same running case — HarbourFront Logistics Pte Ltd and its CustomerConnect portal — so the artefacts learners build compound across the two days. Full step-by-step instructions for each activity are in the Learner Guide.

#	Activity	Tool	Min	LO	TSC	Learner Output
1	Empathise with the Customer to Reframe a Failing Project	Design Thinking	45	LO1	K2, A5	A completed empathy map and persona for Priya Menon, a reframed problem statement, and a ranked list of the top 3 customer-value features with the reasoning for each.
2	Diagnose the Waterfall Failure with a Fishbone Analysis	Fishbone	45	LO1	K1, K3, A2	A completed fishbone diagram with 6 categories and at least 18 causes, each tagged as structural or

						behavioural, plus a one-page recommendation identifying which causes Agile addresses and which it does not.
3	Build the Product Backlog and Run Sprint 1 Planning	Scrum Board	60	LO2	K6, K7, A3	A product backlog of at least 12 estimated user stories in priority order, a Sprint 1 backlog of about 20 points, one written sprint goal, and a Definition of Done agreed by the whole team.
4	Clarify Agile Role Accountability with a RACI Matrix	RACI Matrix	45	LO2	K5, A1, A2	A completed RACI matrix covering 12 activities across 5 roles, a list of the anti-patterns found, and the corrected matrix with the specific change that resolves the developers' escalation.
5	Execute Sprint 1 and Track It on the Scrum Board	Scrum Board	60	LO3	K5, A4, A7	A completed 10-day sprint board, a burndown chart with all 10 data points, an impediment log with the resolution of each of the 3 injections, and the actual velocity achieved.
6	Run the Sprint Retrospective with 5 Whys Root-Cause Analysis	5 Whys	45	LO3	A6, A7	A completed 5 Whys chain of at least 5 levels reaching an actionable root cause, plus one SMART improvement action with a named owner, a measure, and a place in the Sprint 2

						backlog.
7	Prioritise Defect Causes with a Pareto Chart	Pareto Chart	45	LO3	A3, A4, A6	A completed Pareto chart of 8 defect categories with a cumulative line, the identified vital few, and a Sprint 4 improvement commitment with a numeric target and the projected defect reduction.
8	Forecast the Release from Velocity and Read the Agile Metrics	Scrum Board	45	LO3	A4, A6, A7	A velocity-based forecast range in sprints and calendar dates, a release burndown, a written interpretation of the CFD and control chart naming the bottleneck, and a one-paragraph answer to the committee.

5. Assessment Plan

Instrument	Detail
Written Assessment (WA)	Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book. Assesses the underpinning knowledge (K1–K7): the business landscape, customer analysis methods, Agile values and principles, Agile methodologies, change management and team formation models.
Case Study (CS)	Case Study (CS) — an applied business scenario with structured tasks, 1 hour, open book. Assesses applied ability (A1–A7): reading a business scenario, ordering a backlog, interpreting metrics, diagnosing a root cause and recommending action.
Conditions	Open book: slides, Learner Guide and approved course materials only. Individual work. No discussion, no photography, no recording.
Competency Decision	Learners must be assessed Competent in BOTH instruments. A learner assessed Not Yet Competent in either instrument may be re-assessed.
Attendance	A minimum of 75% attendance is required to be eligible for assessment and funding. Learners must be assessed Competent in both instruments.
Appeal	A learner may appeal an assessment decision in writing within 7 working days. The appeal is reviewed by a second qualified assessor.

6. Trainer Delivery Notes

6.1 Timing Discipline

- The activities are the course. If you are running late, compress the concept slides — never the activities. Learners are assessed on applied ability.
- Announce a hard time for each activity and hold it. Activity 5 (the 10-day sprint simulation) will overrun if you do not call each simulated day yourself.
- Keep the debrief for each activity — it is where the learning is consolidated and where the TSC link is made explicit.

6.2 Facilitating the Activities

- Form teams of 3–4 on Day 1 and keep them for both days, so their case artefacts accumulate.
- In Activity 3, you play the Product Owner. Push back on teams that accept every story; make them defend the ordering.
- In Activity 5, inject the three impediments on cue (Day 3 stale feed, Day 5 capacity loss, Day 7 mid-sprint request). The Day 7 request is the real test — watch whether teams protect the sprint or simply absorb the work.
- In Activity 6, do not let a team stop at 'the carrier's fault'. Keep asking why until the cause is something the team itself can change.
- In Activity 8, expect teams to recommend 'increase velocity'. That is the teachable moment — walk them back to the control chart and Little's Law.

6.3 Common Learner Misconceptions to Correct

Misconception	The correction
“Agile means no planning.”	Agile plans continuously and at multiple levels. It replaces one large upfront plan with frequent replanning.
“Agile means no documentation.”	Agile documents 'barely sufficiently' — just in time and for a stated reason. The discipline moves; it does not disappear.
“Velocity measures productivity.”	Velocity is a forecasting input. Rewarding it produces estimate inflation and destroys the forecast.
“Agile gets more done for less money.”	Agile buys responsiveness, earlier value and lower risk of building the wrong thing — not free capacity.
“The Scrum Master runs the team.”	The Scrum Master owns the process and removes impediments. The Developers own how the work is done.
“Working faster reduces cycle time.”	Little's Law: cycle time falls when WIP falls. Starting less finishes more.
“Agile suits every project.”	Fixed regulatory scope, no customer access or physical irreversibility all weigh against it. Say so honestly.

6.4 Contingency

- If a tool URL is unreachable, run the activity on flip-chart paper — every tool here has a paper equivalent, and the thinking is what is assessed.
- If the class is smaller than 6, run two teams and have them cross-review each other's artefacts instead of presenting to the room.

- If a learner arrives without a laptop, pair them into a team that has one; every activity is a team activity.